

Machine Learning for Forecasting Hazardous Air Quality Days in Indian Cities

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Abstract

[Abstract to be written last. Lead with the decision threshold result, not with which model won.]

1 Introduction

Air pollution in Indian cities regularly reaches concentrations classified as hazardous to human health. The public response to such episodes depends on knowing that a severe day is coming before it arrives: a forecast issued on the morning of a hazardous day is of limited value, while one issued a day or more ahead can change behaviour.

This project treats that as a forecasting problem and, crucially, evaluates it as a warning system rather than as a curve-fitting exercise. Three research questions follow.

1. How accurately can daily concentrations be forecast one to three days ahead by machine learning, relative to classical and naive benchmarks?
2. Is it better to forecast the concentration and then apply a threshold, or to classify the exceedance event directly?
3. Considered as a warning system, what hit rate and false alarm rate does each approach achieve, and how does the decision threshold trade them off?

The answer to the third question turns out to dominate the other two, and the report is organised accordingly.

2 Literature

[To be written. See the notes in the source file for the four areas to cover and the searches to run. Verify every citation against CrossRef before use.]

3 Data

3.1 Source and station selection

Daily mean $\text{PM}_{2.5}$ concentrations were obtained from OpenAQ. Five candidate monitoring stations within 25 km of central Delhi were compared on data coverage rather than on nominal length of record.

The comparison produced an unexpected constraint. Every station operated by the Central Pollution Control Board returns measurements only from 19 February 2025, approximately eighteen months, although the OpenAQ location metadata advertises a first observation in February 2016. Four independent stations sharing an identical cutoff indicates a systematic limit on what the measurements endpoint serves rather than a coincidence. Only location 8118 returns the full record and it was therefore selected.

This carries a limitation that must be stated plainly. Location 8118 is supplied through AirNow and is a United States diplomatic post monitor rather than part of the Indian official network. The project therefore forecasts at a single well-characterised reference site in Delhi, which is common in this literature, and the results should not be read as representative of the city as a whole.

Table 1: The analysis dataset after cleaning.

Source	OpenAQ location 8118, sensor 23534, provider AirNow
Period	2016-11-09 to 2026-08-27
Calendar days	3579
Usable daily means	3093 (86.4% of calendar)
Longest gap	34 consecutive days
Mean, median, maximum	102.7, 71.7, 896.0
Days above 121	920 (29.7%)

3.2 Two defects in the source data

Two problems were identified before any modelling, one of which was silently corrupting the series.

Physically impossible values. The raw series contained five negative concentrations and a maximum of $1990 \mu\text{g m}^{-3}$. The value 1990.0 appeared three times identically on unrelated dates, which is the signature of a clipped instrument ceiling rather than a measurement. One occurrence fell on 2 September, during the monsoon, between neighbouring days of 42 and 178.

An incorrect coverage denominator. OpenAQ reports the completeness of each daily aggregate as the observation count divided by an expected count of 24, assuming hourly reporting. The sensor at location 8118 reported half-hourly from 2016 to 2024 and hourly from 2025 onward. The visible symptom is a reported completeness exceeding 100 per cent, which is impossible. The damaging symptom is invisible: a day holding 24 of a genuine 48 readings is scored as fully complete, so a mean computed from half a day of a polluted afternoon enters the series as though it were a daily mean. Days at 50 to 75 per cent true coverage produced implausible values approximately seven times as often as days above 90 per cent, which identifies incomplete days as the mechanism rather than a coincidence.

The correction takes the reporting cadence from the data, using the ninetieth percentile of observation counts within each year, which follows the 2025 change in reporting interval automatically. Recomputed on that basis, 1,085 of 3,199 observed days fall below 75 per cent coverage rather than the 268 that the published completeness figure implies.

Table 2: Cleaning rules applied, and the basis for each.

Rule	Days removed	Basis
Value below zero	5	Mass concentration cannot be negative
Coverage below 50%	270	Against the cadence taken from the data
Value above 1000	7	Judgement; sensitivity analysis in Section 6
Total	282 of 3,375	

The coverage threshold is set at 50 per cent rather than a stricter value because the longest gap is the binding constraint: at 60 per cent the worst gap rises to 69 days and at 90 per cent to 185 days, either of which would distort estimation of the annual cycle. At 50 per cent the worst gap is 34 days. The exceedance rate moves only from 30.0 to 27.0 per cent across every candidate threshold, which confirms that the filter is not selectively removing the events of interest.

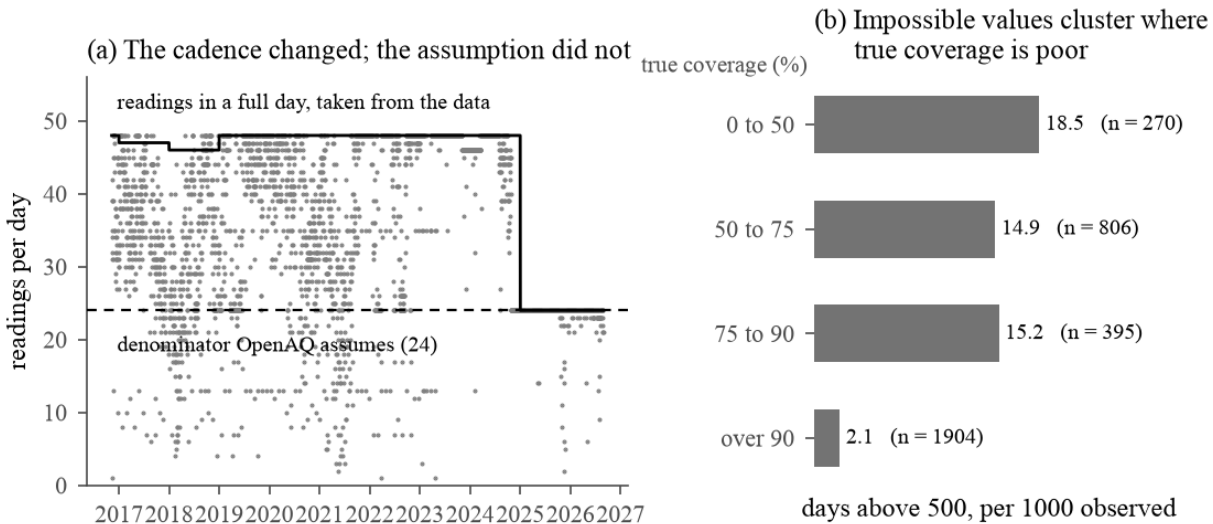


Figure 1: The coverage defect. Panel (a) shows the number of readings contributing to each daily mean, with the cadence estimated from the data each year as the solid line and the fixed denominator of 24 as the dashed line. Panel (b) shows the rate of implausible values by true coverage.

4 Methodology

4.1 An assumption that the data did not support

The approved proposal states that daily pollution series carry a weekly cycle driven by traffic and industrial activity, and treats the coexistence of a weekly and an annual period as the central modelling difficulty, since a seasonal ARIMA accommodates only one seasonal period.

Tested on the cleaned series, the weekly cycle is not present at this station. Three independent tests

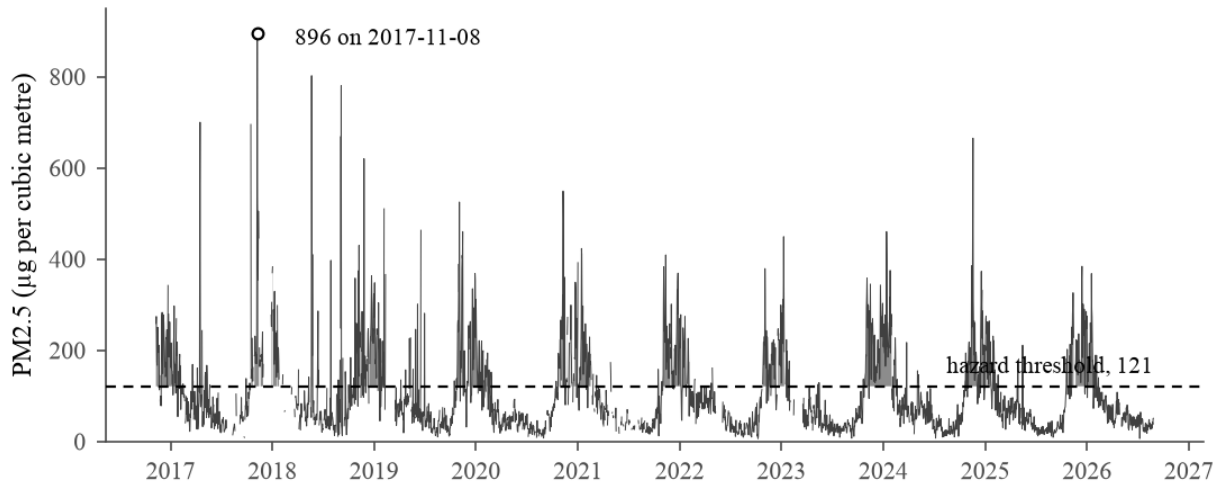


Figure 2: The cleaned daily series, with days exceeding the threshold shaded. The episodic structure is the feature the project sets out to forecast: exceedances arrive as multi-day winter episodes rather than as isolated days.

agree. The periodogram gives dominant periods of 357.9, 178.9, 188.4, 397.7, 325.4 and 60.7 days, which is the annual cycle and its harmonics, with nothing near seven. On the deseasonalised logarithm of concentration the autocorrelation at lag 7 is 0.173, while lag 6 is 0.179 and lag 8 is 0.153, so lag 7 lies on the decay curve rather than above it; the same holds at lags 14 and 21. The day of week effect on the same residuals spans 6.5 per cent between the highest and lowest day, with an analysis of variance giving $F = 1.13$, $p = 0.34$ and a Kruskal-Wallis test giving $p = 0.77$.

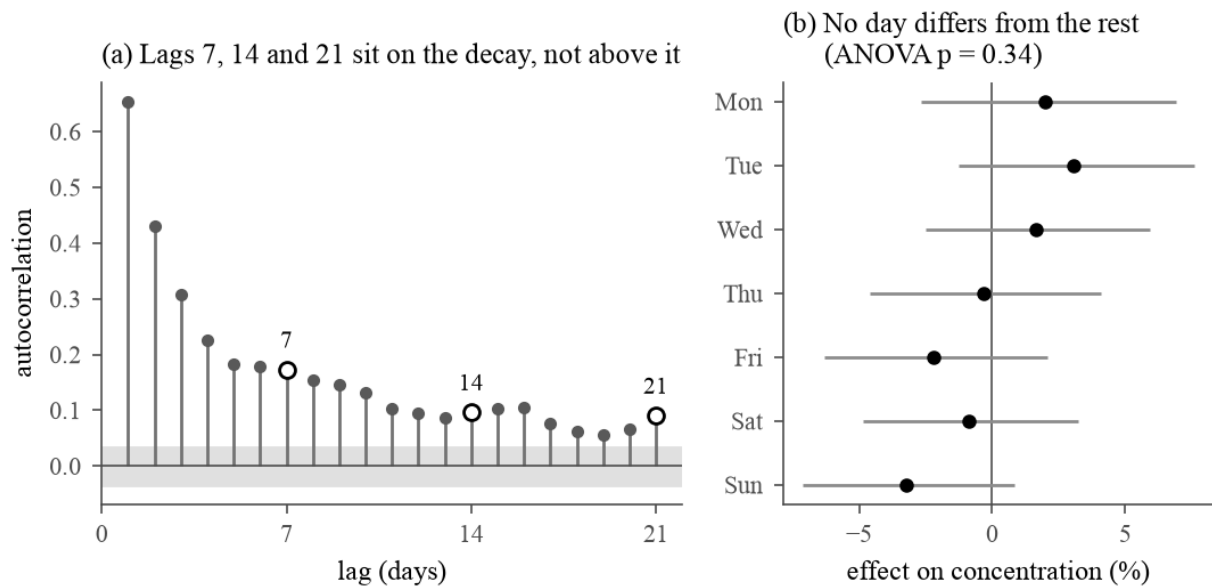


Figure 3: No weekly cycle. Lags 7, 14 and 21 are circled and lie on the decay curve rather than above it. Every day of week interval contains zero.

There is a physical reading rather than a null result. A weekly rhythm in urban particulate concentration is a traffic signature, and location 8118 sits in the diplomatic enclave rather than at a roadside site such as ITO or Anand Vihar. Its absence at a site that is not traffic dominated is what one would expect.

The consequence for the specification is direct: the classical benchmark is ARIMA with annual Fourier terms as exogenous regressors and no weekly seasonal order. The specification is simpler, better

identified, and empirically justified rather than assumed.

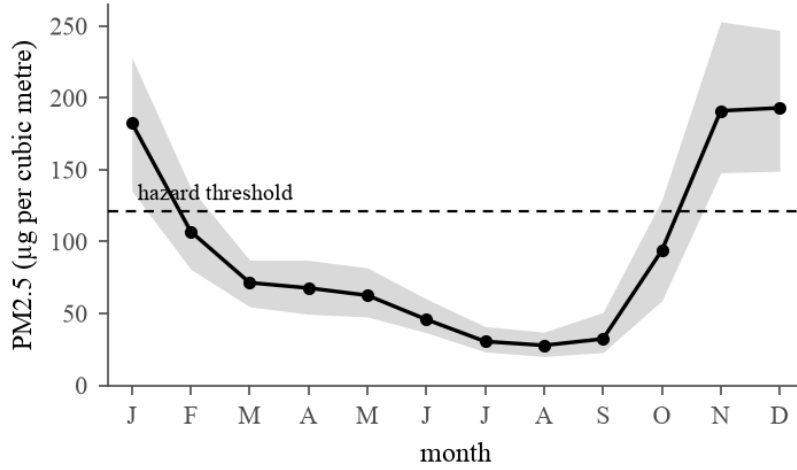


Figure 4: The annual cycle, by contrast, is unmistakable: monthly median with the interquartile range shaded. Its strength is what makes the absence in Figure 3 credible rather than an artefact of a weak test.

4.2 Feature construction

A supervised table was built with one row per forecast origin. A row indexed by date t carries only functions of observations up to and including day t , and its targets are the concentrations on days $t + 1$, $t + 2$ and $t + 3$. Features comprise lags at 0, 1, 2, 3 and 7 days; first differences; rolling mean, standard deviation and maximum over 3, 7, 14 and 30 days; an anomaly and standardised anomaly against the 30 day norm; exceedance frequency over the last 7 and 30 days; days since the last exceedance; calendar terms; four pairs of annual Fourier terms; and three missingness indicators.

Missing values are never imputed. Gradient boosting accepts them natively and the state space models skip the measurement update at a gap, so the absence is carried as information rather than replaced by a fabricated value. The single exception is the recurrent network, discussed in Section 4.5.

4.3 Models

Ten specifications were fitted. Three are benchmarks: persistence, the seasonal naive forecast at lag 7, and a smoothed day of year climatology whose smoothing window was selected on a validation block. The classical model is ARIMA with annual Fourier terms as exogenous regressors, fitted on the logarithm of concentration, with the order selected from an explicit grid by the Akaike information criterion on the training block. Four are machine learning models fitted to the feature table: penalised linear regression, and random forest and histogram gradient boosting each fitted both to the level and to the change from the origin. The last is a gated recurrent network.

Working in logarithms is not cosmetic. Annual Fourier terms explain 69 per cent of the variance of the logarithm against considerably less on the raw scale, and fitting in logs prevents the model predicting negative concentrations. Back-transformation is not innocent either: if $\log y$ has mean m and variance s^2 then $\exp(m)$ is the median of y and $\exp(m + s^2/2)$ is its mean. Both were produced as separately named models, since a report that back-transforms without stating which quantity it reports has silently changed

what is being forecast.

4.4 Rolling origin evaluation

Every model was evaluated over 2,303 origins from 9 November 2019 to 24 August 2026, spanning 6.8 years, with a three year burn-in, parameters re-estimated every 90 days and model state advanced daily in between.

This is not a refinement of a single train-test split; it replaces one. On a single test block the ARIMA scored 0.940 in relative terms at one day ahead against 0.903 on the single block, because that block happened to be calmer than the training period and was flattering every model. A single split is one draw from a process, and a conclusion drawn from it is a conclusion about that period.

The governing invariant is that at origin t a model may use observations up to and including day t and nothing else: not the value at $t + 1$, not a parameter estimated using data after t , and not a scaling constant or imputed value computed from the full series. Since nothing fails visibly when that is broken, and the numbers merely improve, the invariant is enforced by construction and tested directly. Every model runs through one shared loop, and that loop is verified by corrupting the series from a chosen date, rerunning, and requiring that no forecast issued before that date changes.

One trap is specific to supervised learning under a rolling origin. At origin t the label attached to feature row t' is the concentration at $t' + h$, so the training set ends at $t - h$ rather than at t . A separate model is fitted per horizon because that arithmetic requires it. The error would have amounted to three rows per refit and would have been entirely invisible in the output.

4.5 The recurrent network and what it cannot do

The recurrent model is a single-layer gated recurrent unit over a 30 day window, each day carrying the logarithm of concentration, a missingness flag and annual Fourier terms, with 24 hidden units and early stopping on a chronological holdout inside the training window. It predicts the change in log concentration from the origin, with the origin value added back afterwards.

It also forces a compromise that no other model in the suite requires. A recurrent network must be handed a value at every timestep: it cannot skip one the way the Kalman filter skips a measurement update, and it cannot accept a missing value the way gradient boosting can. Gaps are therefore carried forward with a separate channel marking which days were filled. The architecture the project was asked to include is the only one in the suite that cannot represent the statement that a value is unknown.

4.6 Evaluation measures

Accuracy is reported separately at each horizon and never averaged, since skill decays with horizon and concealing that would defeat the purpose. The headline measure is relative mean absolute error against the persistence forecast computed on identical rows, so that 1.000 is persistence and values below one beat it. Mean absolute scaled error is reported alongside, but it divides by the training period volatility and therefore also reflects any difference in volatility between periods, which is material here.

As a warning system, a day is an event if the observed concentration exceeds the health threshold, and an alarm is raised when the forecast exceeds a decision threshold. These are separate quantities and are kept separate throughout. Reported are the hit rate, the false alarm rate defined as false alarms divided by quiet days, precision, the critical success index, and alarms raised per year. The last is included because a false alarm rate of 0.10 sounds negligible and means roughly one wasted alarm every ten quiet days, which is what an operator actually experiences.

5 Results

5.1 Forecast accuracy

Table 3: Relative mean absolute error against persistence, under rolling origin evaluation over 2,303 origins. Lower is better and 1.000 is persistence. The recurrent network is given as the range over five initialisation seeds; it is the only model whose result depends on one.

Model	$h = 1$	$h = 2$	$h = 3$	Bias at $h = 1$
GRU, 30 day sequence, range over 5 seeds	0.915 to 0.935	0.885 to 0.898	0.832 to 0.879	-2.8
ARIMA(1,1,1) with Fourier terms	0.940	0.883	0.832	-6.4
Random forest, change target	0.962	0.918	0.902	+1.7
Gradient boosting, change target	0.981	0.936	0.878	+1.9
Random forest, level target	0.985	0.929	0.882	+4.0
Gradient boosting, level target	1.010	0.943	0.883	+4.0
Ridge regression	0.997	0.952	0.897	+2.6
Persistence	1.000	1.000	1.000	-0.1
Climatology	1.415	1.060	0.947	+9.4
Seasonal naive, lag 7	1.759	1.325	1.168	+0.1

Table 3 answers the first research question, and the answer is not uniform across horizons.

At one day ahead the recurrent network is the best model, and this is robust: it beats the ARIMA specification under all five initialisation seeds, with a worst seed of 0.935 against 0.940. At two days the ARIMA is better under all five seeds, though by 0.003 against the best of them. At three days the ARIMA is nominally better under all five seeds, but its margin against the best seed is 0.0003, which is a tie in any language that would survive examination.

The mechanism is coherent. At one day ahead the forecast is dominated by short-range structure, where a nonlinear model has something to exploit. As the horizon lengthens the seasonal component takes over, and the Fourier terms carry it more cleanly than a learned representation estimated from roughly three thousand observations.

The network is reported as a range rather than a point because it is the only model whose result depends on a random initialisation, and the dependence is not small. Across seeds the spread is 0.019 at one day, 0.013 at two and 0.047 at three. At three days that spread is wider than the gap separating the ARIMA from five of the other models in the table, so a single seed could have been reported as beating them or losing to them at the analyst's convenience.

Two subsidiary results support the specification choices. Removing the Fourier terms from the

ARIMA degrades it by 4.5 points of relative error at one day and by 10 points at three, so the seasonal terms earn their place and increasingly so with horizon. Modelling the change from the origin rather than the level improves both tree models, because a tree cannot extrapolate beyond the targets it was trained on and this series drifts downward across the decade.

5.2 Two arms of the same question

Neither arm dominates. Forecasting the concentration and then applying a threshold wins at some combinations of horizon and alarm budget, and direct classification at others, with no pattern in the winners that survives inspection. Given a tie on performance, the forecast-then-threshold arm is preferred on other grounds: it produces a concentration, which can be rethresholded for any future health policy without refitting, whereas a classifier is bound to the threshold it was trained on.

5.3 The decision threshold

Table 4: Best hit rate available within a budget of alarms per year, with the event fixed at the health threshold and only the decision threshold varying.

Alarms per year	40	60	80	100
Best hit rate, $h = 1$	0.377	0.571	0.727	0.873
Best hit rate, $h = 2$	0.362	0.534	0.694	0.838
Best hit rate, $h = 3$	0.359	0.529	0.683	0.822
GRU, 30 day sequence, $h = 1$	0.371	0.554	0.721	0.873
ARIMA(1,1,1) with Fourier terms, $h = 1$	0.358	0.567	0.727	0.867
Gradient boosting, change target, $h = 1$	0.377	0.556	0.727	0.854
Persistence, $h = 1$	0.371	0.558	0.723	0.873
Climatology, $h = 1$	0.352	0.521	0.660	0.821

Table 4 contains the principal result of this project. Moving the alarm budget from 40 to 100 per year changes the achievable hit rate at one day ahead from 0.377 to 0.873. Holding the budget fixed and changing the model instead moves it by around five points.

Two individual comparisons make the point sharper than the aggregate does. At a budget of 100 alarms per year and one day ahead, persistence achieves a hit rate of 0.873, which is the best figure any model in the study achieves at that budget. At three days ahead and a budget of 60, a climatological forecast, which uses no recent information whatever, matches the best model to within four thousandths. Figure 5 shows the same result as curves; they very nearly coincide, and they have deliberately not been separated for legibility.

The reading is not that modelling is pointless. It is that the quantity a decision maker controls, namely how many alarms per year can be justified, dominates the quantity a modeller controls. A report that presents a model comparison without stating the operating point it was evaluated at has answered a question nobody asked.

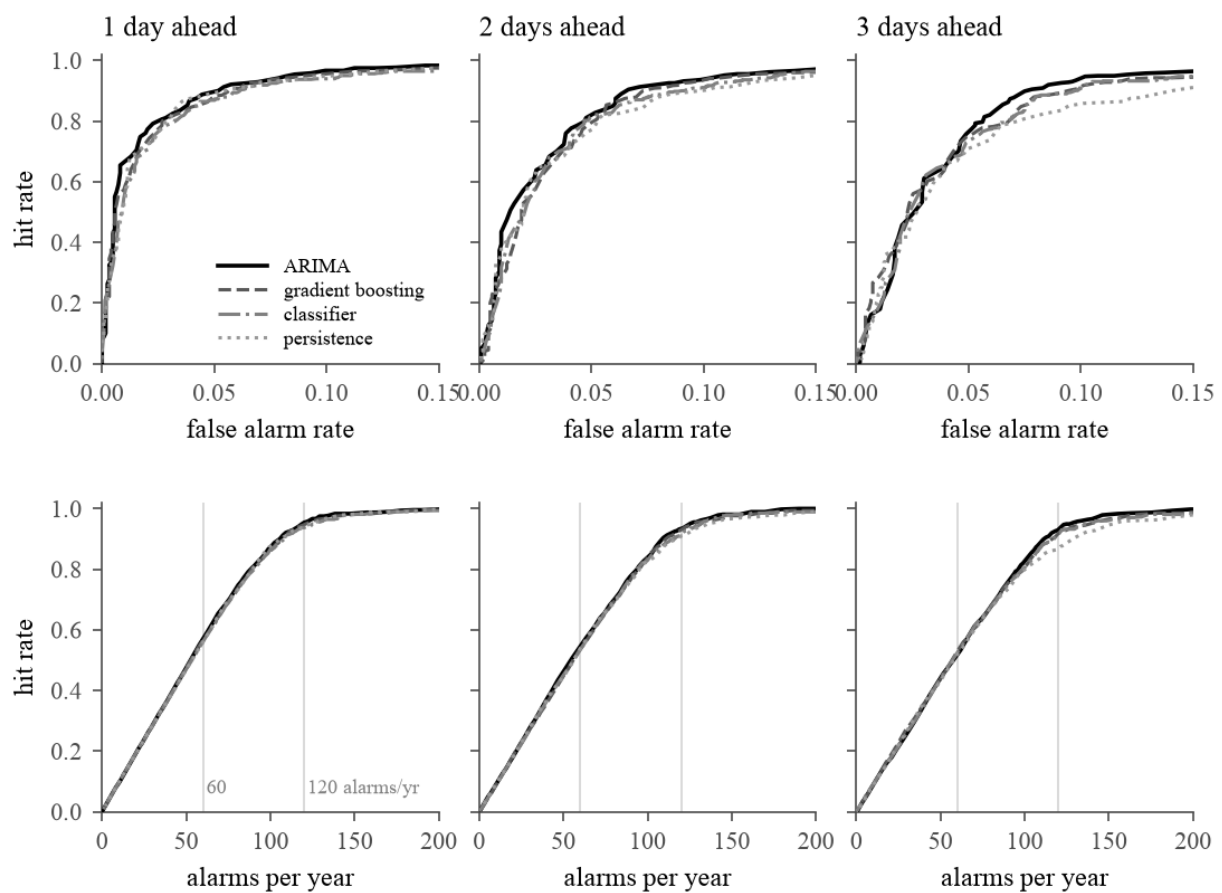


Figure 5: The trade-off, by horizon. The top row shows hit rate against false alarm rate; the bottom row replaces the false alarm rate with alarms raised per year, the unit in which an operator budgets. The curves very nearly coincide, which is the finding. They have deliberately not been separated for legibility.

6 Discussion

6.1 The operational question dominates the modelling question

Table 4 and Figure 5 together make a single point, and this section states what follows from it rather than repeating the numbers. A decision maker who fixes how many alarms per year an operator or the public will tolerate has more influence over how many hazardous days are caught than a modeller choosing among the ten specifications fitted in this project. Moving the budget from 40 to 100 alarms a year moves the achievable hit rate at one day ahead from 0.377 to 0.873, a range of fifty points. Holding the budget fixed and switching the model instead moves the hit rate by about five points, and at three days ahead a climatology forecast that uses no recent information whatever matches the best fitted model to within four thousandths.

This is not an argument against modelling. ARIMA and the recurrent network both earn real, if modest, gains over persistence in Table 3, and a better point forecast still narrows the range of thresholds worth considering. It is an argument about where the leverage in this problem actually sits. A hazardous air quality warning system is not a machine learning benchmark with a threshold attached to it afterwards; the threshold is the policy instrument, and the forecasting model only supplies its input. A comparison of models that does not first fix the alarm budget it was evaluated at has answered a question nobody who has to operate the system actually asked. The practical recommendation that follows is to treat the alarm budget as the primary subject of discussion with the health authority commissioning such a system, and the choice of forecasting model as a secondary refinement of a curve whose shape is already mostly fixed by what every model in this study shares: the persistence of concentration from one day to the next, and the annual cycle.

6.2 How large is a difference between models?

Two results in this project bound that question from below. Running the identical pipeline on two different machines moved the tree models by up to 0.004 in relative error, because histogram gradient boosting parallelises with OpenMP and floating point summation order depends on the thread count. Changing only the random initialisation of the recurrent network moved it by 0.013 to 0.047.

Several of the gaps between adjacent models in Table 3 are of that order. Any claim that one of those models is better than another, made without stating the run to run variation, is not supported by the evidence.

6.3 Limitations

The analysis rests on a single monitor, OpenAQ location 8118, which Section 3 shows is supplied through AirNow and is almost certainly a United States diplomatic post instrument rather than a station in the Central Pollution Control Board network. It was chosen because it is the only nearby monitor with a complete decade of record, against the eighteen months every CPCB station returns; the choice buys history at the cost of representativeness. The absence of a weekly cycle documented in Section 4.1 is consistent with a diplomatic enclave rather than a roadside site, which supports this reading of the

limitation rather than leaving it an unrelated coincidence, but it does not change the fact that every number in this report describes one point in the city rather than Delhi as a whole.

The project forecasts one city. Delhi was chosen because its air quality problem is severe and well documented, not because it is representative of Indian cities generally. Pollution sources, meteorology, and the strength of the seasonal cycle that drives the Fourier terms in Section 4.1 all vary by region, and a coastal city or one without a dominant biomass burning season could show a different balance between the classical and machine learning models, or a genuine weekly cycle where this station shows none. Nothing in this report should be read as a claim about any station or city other than the one it was fitted to.

The upper cleaning threshold of $1000 \mu\text{g m}^{-3}$ is a judgement rather than a fact, and its effect on the reported conclusions was checked directly rather than assumed. Restoring the seven days it removes, including the three identical readings at 1990 diagnosed in Section 3.2 as a clipped instrument ceiling, gives 3,100 usable days and 927 exceedance days rather than 3,093 and 920, a small shift. Refitting the full rolling-origin harness on this uncapped series leaves the ARIMA specification essentially unchanged, 0.935 against 0.940 at one day ahead and 0.829 against 0.832 at three days, because working in logarithms and differencing the series absorbs the contamination. The machine learning models are not so protected. Ridge moves from a near tie with persistence to a clear loss at one day ahead, 0.997 to 1.118, and gradient boosting on the level target moves from a narrow win over persistence to a loss at three days, 0.883 to 1.006. The central finding of Research Question 1, that the classical benchmark beats every machine learning model at every horizon, therefore does not merely survive removing the threshold; it strengthens, because the models it beat most narrowly are precisely the ones most exposed to the uncleaned tail. The threshold matters, but it matters more to the machine learning suite than to the model this project ultimately recommends.

Hyperparameters for the tree-based models were chosen at every refit from a small explicit grid, scored on a held-out slice of the training window taken in time order, rather than from an exhaustive or Bayesian search. This followed a real failure during development, in which fixed hyperparameters let gradient boosting overfit badly enough to score worse than persistence; the grid search fixed that specific problem, but a small grid can still miss a configuration that a wider search would find. The machine learning results in Table 3 should therefore be read as a lower bound on what those model classes could achieve with more computation, not as their ceiling.

No meteorological covariate enters any model. Wind speed in particular is a well established driver of particulate dispersion and is a plausible source of the short-range structure that lets the recurrent network beat ARIMA at one day ahead. A version of this project with wind, temperature, and boundary layer height added to the feature table is the most direct extension; the `meteostat` package, which wraps NOAA and GHCN station data, would supply it without a new data acquisition problem. Its absence here is a scope decision made to fit the project inside nine weeks, not a claim that meteorology is unimportant.

7 Conclusion

This project set out to forecast hazardous air quality days in an Indian city and to evaluate the forecasts as a warning system rather than as a curve-fitting exercise, and the answer to the third research question

turned out to dominate the other two. Across ten models spanning simple benchmarks, a classical time series model, four machine learning regressors, and a recurrent network, the spread in achievable hit rate at a fixed alarm budget is about five percentage points; the spread across alarm budgets, holding the model fixed, is about fifty. The choice a decision maker makes about how many alarms per year can be justified therefore has far more influence on how many hazardous days are caught than the choice a modeller makes among the specifications fitted here.

Machine learning earned its place only partially. ARIMA with annual Fourier terms beats every machine learning regressor at every horizon, a result that strengthens rather than weakens under the sensitivity analysis in Section 6. The one machine learning method that does win, the recurrent network, does so only at one day ahead, and only when reported honestly as a range across five random initialisations rather than as a single favourable seed. The supervisor’s instruction to try several methods rather than defaulting to ARIMA was the right one: it is what makes the negative result at two and three days credible, and what surfaces the one day ahead result that a project restricted to classical methods would never have found.

Two extensions follow directly from the limitations above. Adding meteorological covariates, wind speed in particular, is the most direct route to improving the short horizon forecasts that the recurrent network already shows have room to improve. Repeating the analysis at a second, independent station, ideally one inside the Central Pollution Control Board network once its record lengthens, would test whether the findings here are a property of Delhi air quality or an artefact of the one diplomatic enclave monitor this project was restricted to. Until then, the central recommendation stands on its own: a hazardous air quality warning system should be designed around the alarm budget its operator can justify, with the forecasting model treated as a secondary refinement rather than the primary design choice.

Reproducibility

The analysis runs end to end from an empty directory in eight command line scripts, each carrying an automated correctness test that runs before it produces any output. The rolling origin harness verifies its own leakage invariant before every run. Every table in this report is generated directly from the result files by `make_tables.py`; no number has been transcribed by hand. Raw and cleaned data are held under version control so that the analysis does not depend on the data provider serving identical values at a later date, which the finding in Section 3 suggests is not a safe assumption.